

Allan Zhang

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Education

University of California, Los Angeles *BS in Computational Mathematics* *Sept 2024 – June 2028*
◦ **Relevant Coursework:** Multivariable Calculus, Linear Algebra, Differential Equations, Discrete Structures, Probability Theory, Data Structures and Algorithms, Large Scale Machine Learning (graduate coursework)

Experience

Research Intern | DatologyAI *Jan 2026 – April 2026*
◦ Built a reinforcement learning framework for VLMs from scratch to generate and evaluate high-quality reasoning traces; optimized weight synchronization between inference workers and trainers
◦ Developed checkpoint resumption system with optimizer state loading for VLM training, enabling fault-tolerant recovery and flexible CPT launch configurations
◦ Created an end-to-end pipeline for filtering synthetic VLM reasoning traces, leveraging Flyte for orchestration, Ray for distributed compute, and Spark for data processing

Undergraduate Research Assistant | BigML @UCLA *Nov 2024 – Present*
◦ Conducting research on data-efficient training for LLMs, focusing on data selection for reinforcement learning and synthetic data generation. Advised by Prof. Baharan Mirzasoleiman
◦ Co-designed a data-efficient variant of GRPO, achieving 30% faster convergence and a 40% reduction in required training samples relative to vanilla GRPO

Projects

Torchure (Pure PyTorch Training Stack) [torchure](#) 🔗
◦ Built an LLM pretraining stack in pure PyTorch: Qwen3 0.6B (GQA, RoPE, SwiGLU), streaming dataloader with sequence packing, CUDA prefetch, and stateful checkpointing
◦ Lifted single-GPU throughput 52% (35.8k → 54.5k tok/s) and cut peak memory 20% with per-block torch.compile and a fused linear + cross-entropy kernel that never materializes 152k-vocab logits
◦ Implemented DDP from scratch: bucketed gradient all-reduce overlapped with backward via post-accumulation hooks, achieved 198k tok/s on 4×H100 at 92% scaling efficiency

Thinking Constrained GRPO [grpo-thinking-budget](#) 🔗
◦ Implemented constrained reasoning for LLMs by enforcing thinking token budgets during GRPO training, enabling fine-grained control over model inference costs
◦ Designed a two-phase generation system with vLLM that separates thinking and response phases, with automatic delimiter injection when thinking budget is exhausted
◦ Created fast inference pipeline using vLLM with two-phase generation to control thinking vs. response token allocation for evaluations

Publications

Beyond What Seems Necessary: Hidden Gains from Scaling Training-Time Reasoning Length under Outcome Supervision 🔗 *Under review, NeurIPS 2026*
Yihao Xue, Allan Zhang, Jianhao Huang, Amit Sahai, Baharan Mirzasoleiman

20/20 Vision Language Models: A Prescription for Better VLMs through Data Curation Alone 🔗 *Technical report, 2026*
DatologyAI

Skills

Programming Languages and Frameworks: Python, PyTorch, JAX, NumPy, Matplotlib, TensorFlow, scikit-learn, OpenCV, Hugging Face, L^AT_EX, C++, Bash, vLLM, SGLang